

IKK Notation Registry

Inverse Kaluza-Klein Scale Symmetry

Symbol Status, Collision Tracking, and Promotion Log

GREEN	Solid / accepted	Use freely
YELLOW	Caution / needs clarification	Define before expanding
ORANGE	Conflict / collision	Rename, type, or namespace
RED	Phasing out / deprecated	Replace in new writing
BLUE	Provisional / newly introduced	Test before promoting

Covers: Doc I (Schrödinger Alternative), Doc II (Initial Draft — Axioms A–J),
Doc III (Kernel V4), Doc IV (5D Interval and Projection Mechanics),
and all future IKK / VSEPR-SIM documents.

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§1 Status Legend

Tag	Name	Meaning	Action
GREEN	Solid / Accepted	Stable, defined, safe.	Use in papers, diagrams, summaries.
YELLOW	Caution / Needs Clarification	Useful but ambiguous.	Define tightly before expanding use.
ORANGE	Conflict / Collision	Collides with other symbol or convention.	Rename, add subscript, or namespace.
RED	Deprecated / Legacy	Too ambiguous; replaced.	Do not use in new writing.
BLUE	Provisional / New	Newly introduced; untested.	Test against full system before promoting.

Notation layer hierarchy. Three tiers govern which font a symbol uses:

- **Primitive / theoretical layer** — Fraktur: $\mathfrak{I}, \mathfrak{D}, \mathfrak{B}, \mathfrak{A}, \mathfrak{R}, \mathfrak{K}$
- **Operational / projection layer** — Calligraphic: $\mathcal{P}, \mathcal{C}, \mathcal{R}, \mathcal{M}, \mathcal{O}, \mathcal{U}$
- **Empirical / data layer** — Bold or blackboard bold: $\mathbb{D}, \mathbf{x}, \mathbf{v}, \mathbf{F}, \mathbf{T}_{AB}$

§2 Core Accepted Symbols

Symbol	Meaning	Status	Notes
$\mathfrak{I}$	Primitive / true identity content	GREEN	Fraktur layer. Never equals $\mathfrak{I}_{\text{top}}$ or $\mathcal{O}$.
$\mathfrak{D}$	Distinguishability / recoverable identity information	GREEN	Primary information variable. Replaces bare I or D .
P	Particle / identity-bearing object	GREEN	Roman P . Never use for projection; that is $\mathcal{P}$.
S_n	Scale level	GREEN	See Sec. 4. Core triple.
$\mathcal{P}_{a \rightarrow b}$	Projection operator	GREEN	Calligraphic only. Canonical form.
$\mathcal{O}_b^{(a)}$	Observable at S_b of identity from S_a	GREEN	Clean projection output symbol.

Canonical identity equation:

$$\mathfrak{I}_{\text{true}} \neq \mathfrak{I}_{\text{top}} \neq \mathcal{O}_b^{(a)}$$

Canonical projection equation:

$$\mathcal{P}_{a \rightarrow b}[\mathfrak{I}_a] = \mathcal{R}_b(\mathcal{C}_{ab}[\mathfrak{I}_a]) = \mathcal{O}_b^{(a)}$$

§3 Projection and Resolution Symbols

Symbol	Meaning	Status	Notes
$\mathcal{P}_{a \rightarrow b}$	Projection operator from S_a resolved at S_b	GREEN	Core operator. Defined in Doc IV Eq. (6.1).
$\mathcal{C}_{ab}$	Cross-scale coupling rule	GREEN	Gives projection physical content.
$\mathcal{R}_b$	Resolution operator of observing scale S_b	GREEN	Separates access from resolution.
$\mathcal{O}_b^{(a)}$	Observable at S_b of identity from S_a	GREEN	Projection output.
$\mathcal{M}_b$	Measurement map	YELLOW	Useful but not always needed. Define role vs. $\mathcal{R}_b$.
χ_{ab}	Scale compatibility / access condition	BLUE	Candidate. Whether projection $a \rightarrow b$ is permitted.
η_{ab}	Projection efficiency / coupling strength	BLUE	Candidate. Partial access weight.
ϵ_{ab}	Projection error / mismatch	BLUE	Candidate. Formalise loss term.
$\Pi_{m \leftarrow n}$	Scale projection operator (Doc IV)	GREEN	Arrow notation from Doc IV. Equivalent to $\mathcal{P}_{n \rightarrow m}$.
Canonical form: $\mathcal{P}_{a \rightarrow b} = \mathcal{M}_b \circ \mathcal{R}_b \circ \mathcal{C}_{ab}$			
Simple form: $\mathcal{P}_{a \rightarrow b} = \mathcal{R}_b \circ \mathcal{C}_{ab}$			

§4 Scale Ladder Symbols

Symbol	Meaning	Status	Notes
S_n	Scale level n	GREEN	Central object.
S_a, S_b	Source / target scale labels	GREEN	Use $a \rightarrow b$ for cross-scale direction.
$a \rightarrow b$	Cross-scale resolution direction	GREEN	Means “resolved from a into b ”; not transport.
D_n	Dimensional degree of freedom at S_n	GREEN	Do not confuse with $\mathfrak{D}$ (distinguishability).
M_n	Mediator / matrix / metric entry	YELLOW	Currently ambiguous. Choose one meaning per document.
L_n	Characteristic length scale	GREEN	Stable. Core scale parameter.
ℓ_n	Local / measured length	YELLOW	Use when distinct from L_n .
λ_n	Scale multiplier / wavelength	YELLOW	Collision with wave mechanics. Use λ_{scale} if needed.
ΔS	Scale separation	YELLOW	Define as $ S_b - S_a $ or log ratio.
w	Fifth / scale coordinate	YELLOW	Doc-specific. Superseded by $u = \ln(\ell/\ell_0)$ in Doc IV.
u	Logarithmic scale coordinate	GREEN	$u = \ln(\ell/\ell_0)$, $du = d\ell/\ell$. Defined in Doc IV.

Core triple: $S_n = (D_n, M_n, L_n)$

Symbol	Meaning	Status	Notes
<i>Warning:</i> M_n is currently wearing three fake moustaches (mediator / matrix / metric). Resolve before use in any new section.			

§5 Identity-Layer Symbols

Symbol	Meaning	Status	Notes
$\mathfrak{I}_{\text{true}}$	Full primitive true identity	GREEN	Deepest identity. Never observable directly.
$\mathfrak{I}_n$	Identity at scale S_n	GREEN	Native-scale identity.
$\mathfrak{I}_{\text{top}}$	Top-layer identity	GREEN	Highest-scale packed identity. $\neq \mathfrak{I}_{\text{true}}$.
$\mathfrak{B}$	Identity bundle	GREEN	Packed / coexisting multi-scale identity object.
$\mathfrak{A}$	Identity anchor	YELLOW	Fixed reference point in identity space. Role needs formalising.
r_F	Fuzz radius	YELLOW	Use only where scale fuzz is formally defined.
P_n	Identity object at scale (Doc IV form)	GREEN	$P_n = (x^\mu, \mathbb{C}_p, D_n, M_n, L_n)$.
$\mathbb{C}_p$	Complex fibre above point p	GREEN	Introduced Doc IV. Gives local phase structure.

Key rule: $\mathfrak{I}_{\text{true}} \neq \mathfrak{I}_{\text{top}} \neq \mathcal{O}_b^{(a)}$

This inequality should appear early in any new paper. Readers will conflate these three unless corrected before they have time to form opinions.

§6 Matrix-Wave Bridge Symbols

Symbol	Meaning	Status	Notes
ψ	Wavefunction	GREEN	Projection-side object. Not the primitive identity.
Ψ	Full wave state / state vector	GREEN	Standard quantum comparison symbol.
$\hat{H}$	Hamiltonian operator	GREEN	Always hatted.
$\hat{\rho}$	Density operator	GREEN	Hatted to distinguish from ρ_m (mass density).
$\mathcal{U}$	Evolution operator	GREEN	$\mathcal{U}(t) = e^{-i\hat{H}t/\hbar}$.
M	Matrix-state representation	YELLOW	Define whether true state or representational.
H	Bare Hamiltonian / HGST / field driver	ORANGE	Three meanings. Use $\hat{H}$, H_{HGST} , or H_∇ .
ρ	Density / density matrix / donor quality	ORANGE	Type every instance: $\hat{\rho}$, ρ_m , D_ρ .

Symbol	Meaning	Status	Notes
Bridge statement: $\psi_b^{(a)} \sim \mathcal{P}_{a \rightarrow b}[\mathfrak{I}_a]$			
Scale-dependent form (Doc IV): $\psi(x, t; L) = \langle x, L \mathfrak{I}_{\text{true}} \rangle$			
ψ is not the primitive identity. It is the projected / resolved form at a chosen scale and position.			

§7 Relation and Entanglement Symbols

Symbol	Meaning	Status	Notes
$\mathbf{T}_{AB}$	Trinary relation vector between A and B	GREEN	Values in $\{-1, 0, +1\}^m$. Doc II Eq. (G.2).
$\mathbf{R}_{AB}$	Primitive relation-state	GREEN	Full relation object.
$[0, 0, 0, 0 \mathbf{T}_{AB}]$	Zero spacetime displacement + relation vector	GREEN	Very distinctive. Use to keep entanglement away from signal language.
Γ_{AB}	Relation / coupling graph	BLUE	Candidate. For graph-identity formalism.
κ_{AB}	Coupling strength	BLUE	Candidate. Interaction / access strength scalar.
Canonical form: $\mathbf{R}_{AB} = [0, 0, 0, 0 \mathbf{T}_{AB}]$			
Entanglement is a real relation-state with zero ordinary spacetime displacement. It is not a signal. It is not mystical. It is a structure.			

§8 Information and Entropy Symbols

Symbol	Meaning	Status	Notes
$\mathfrak{D}$	Distinguishability / recoverable identity content	GREEN	Primary information symbol in the theory.
S	Entropy	GREEN	Use S for thermodynamic entropy. Not bare S (scale collision).
ΔS	Entropy change	GREEN	Standard.
$\Delta \mathfrak{D}$	Distinguishability change	GREEN	Useful for SM's law.
$\mathfrak{L}_D$	Information-loss functional	BLUE	Candidate. Cleaner than vague "loss."
Λ_D	Information-loss operator	BLUE	Optional. Useful if formalising dynamics.
μ_D	Mean distinguishability	BLUE	Optional. Statistical sections.
σ_D	Distinguishability variance	BLUE	Optional. Measurement uncertainty sections.
I	Information / identity (legacy)	RED	Retire. Replace with $\mathfrak{I}$ or $\mathfrak{D}$.

Symbol	Meaning	Status	Notes
S	Scale / entropy / steric penalty (bare)	ORANGE	Three meanings. Use S_n , S_{ent} , S_{steric} .
Core relation: $\Delta S \propto -\Delta \mathfrak{D}_{\text{rec}}$			
$\Delta S > 0 \iff \Delta \mathfrak{D} < 0$			
Entropy is loss of recoverable distinguishability. This is the IKK restatement of the second law.			

§9 Particle-by-5 and State Symbols

Symbol	Meaning	Status	Notes
P_i	Particle / identity-bearing object	GREEN	Core.
x_i	Position	GREEN	Standard.
p_i	Momentum	GREEN	Preferred over v_i in theoretical sections.
v_i	Velocity	GREEN	Use in simulation / VSIM sections.
m_i	Mass	GREEN	Standard.
q_i	Charge	GREEN	Standard.
χ_i	Internal identity state	YELLOW	Candidate replacement for s_i . Better when spin is also present.
s_i	Spin / state / scale	ORANGE	Too overloaded. Prefer χ_i for internal state.
θ_i	Phase / orientation	YELLOW	Type by context: θ_{phase} , θ_{orient} .
ω_i	Angular frequency / angular velocity	GREEN	Standard.
Q_i	Charge / state matrix	ORANGE	Also used in HGST state vector. Type subscript.
Theory form: $P_i = [x_i, p_i, q_i, \mathfrak{D}_i, \mathfrak{I}_i]$			
Simulation form: $P_i = [x_i, v_i, m_i, q_i, \chi_i]$			

§10 Force and Field Symbols

Symbol	Meaning	Status	Notes
F_i	Total force on object i	GREEN	Standard.
$F_{i,\text{bond}}$	Bonded force	GREEN	Stable.
$F_{i,\text{near}}$	Near-field force	GREEN	Stable.
$F_{i,\text{far}}$	Far-field force	GREEN	Stable.
$F_{i,\text{flux}}$	Flux-mediated force	GREEN	Stable.

Symbol	Meaning	Status	Notes
Φ_k	Field potential k	GREEN	Good.
$\nabla\Phi_k$	Field gradient	GREEN	Conventional and clean.
$\mathbf{B}$	Magnetic field	GREEN	Bold $\mathbf{B}$. Reserve for magnetism only.
$\mathfrak{B}$	Identity bundle	GREEN	Fraktur $\mathfrak{B}$. Does not collide with $\mathbf{B}$.
H	Gradient driver / field driver	ORANGE	Collides with Hamiltonian. Use H_∇ or $\hat{H}$.
Force split: $F_i = F_{i,\text{bond}} + F_{i,\text{near}} + F_{i,\text{far}} + F_{i,\text{flux}}$			

§11 Geometry and Fifth-Dimension Symbols

Symbol	Meaning	Status	Notes
x^μ	Spacetime 4-vector	GREEN	Standard.
ds_4^2	4D spacetime interval	GREEN	Standard SR form.
ds_5^2	5D scale-extended interval	GREEN	Defined Doc IV. $= -c^2 dT^2 + dr^2 + \eta_u (d\ell/\ell)^2$.
u	Logarithmic scale coordinate	GREEN	$u = \ln(\ell/\ell_0)$.
η_u	Scale-axis metric coefficient	YELLOW	Free parameter. Needs experimental constraint.
$a_i(\ell)$	Scale-modulated metric coefficient	GREEN	$a_i \in [0, 1]$. Defined Doc IV Eq. (3.2).
$D_{\text{eff}}(\ell)$	Effective dimension at scale ℓ	GREEN	$= \sum_i a_i(\ell) + a_u(\ell)$.
$g_{\mu\nu}$	Metric tensor	GREEN	Standard.
ℓ_0	Reference length scale	GREEN	Fixed normalisation for u .
$A_Q(\ell)$	Property activation function	GREEN	Sigmoid in log-scale. Doc IV Eq. (8.2).
L_Q	Activation length for property Q	GREEN	Inflection point of A_Q .
k_Q	Sharpness of activation	YELLOW	Free parameter per property. Needs calibration.
χ_w	Dark-scale proper-time correction	YELLOW	Partially identified in Doc IV Eq. (2.6). Needs η_u to close.
w	Old fifth-coordinate label	RED	Replace with u in new writing.
5D interval: $ds_5^2 = -c^2 dT^2 + dr^2 + \eta_u \left(\frac{d\ell}{\ell}\right)^2$			

§12 Project-Specific / VSEPR-SIM Symbols

These symbols belong to the VSEPR-SIM simulation namespace. Do not import them into pure theory sections without typed subscripts.

Symbol	Meaning	Status	Notes
$\mathbf{x}_{\text{chem}}$	Chemical state vector	GREEN	= [D_ρ , Γ_{geom} , S_{steric} , Π_{ox} , Q_{state}]
D_ρ	Donor coordination quality	GREEN	Formerly ρ_D (ambiguous).
Γ_{geom}	VSEPR geometry / symmetry score	GREEN	Fine in chemistry context.
S_{steric}	Steric penalty	GREEN	Typed to avoid collision with S_n and S_{ent} .
Π_{ox}	Oxidation plausibility	GREEN	Π reserved for chemistry context only.
Q_{state}	Charge / oxidation state quality	GREEN	Typed.
H_{HGST}	HGST transform matrix	GREEN	Always typed.
$\bar{\eta}$	Average packing / coordination metric	YELLOW	Project-side use only.
S	Steric penalty (bare)	RED	Replace with S_{steric} .
Π	Oxidation plausibility (bare)	ORANGE	Do not use for projection. Π_{ox} only.

§13 Operator and Transformation Symbols

Symbol	Meaning	Status	Notes
$\hat{A}$	Generic quantum operator	GREEN	Hatted. Standard.
$\mathcal{U}$	Evolution operator	GREEN	$\mathcal{U}(t) = e^{-i\hat{H}t/\hbar}$. Standard quantum-friendly.
$\mathfrak{K}$	Primitive kernel operator	GREEN	Fraktur layer. Foundation-level.
K	Kernel operator (operational)	GREEN	Calligraphic or roman K depending on context.
$\mathcal{M}_b$	Measurement map at S_b	YELLOW	Useful; needs defined role vs. $\mathcal{R}_b$.
T	Transformation / time ordering	YELLOW	Could collide with $\mathbf{T}_{AB}$. Use $\hat{T}$ for time ordering.
E	Evolution / energy	ORANGE	Use $\mathcal{U}$ for evolution, E only for energy eigenvalues.
G	Gate / geometry / graph	ORANGE	Define explicitly by context.
N	Noise / nullspace / normalisation	YELLOW	Needs explicit meaning in each section.

§14 Deprecated and Legacy Symbols

All symbols in this section carry status **RED**. Do not use in new writing. Quote in legacy context with explicit deprecation note.

Deprecated	Replace with	Reason	First deprecated
I (plain)	$\mathfrak{I}$ or $\mathfrak{D}$	Identity / information / current collision.	Doc IV, 2026-05-24
P (projection)	$\mathcal{P}_{a \rightarrow b}$	Collides with particle P and probability.	Doc IV, 2026-05-24
O (plain)	$\mathcal{O}_b^{(a)}$	Too plain; unreadable in equations.	Doc IV, 2026-05-24
S (bare scale)	S_n	Collides with entropy and steric penalty.	Doc II
H (bare)	$\hat{H}$, H_{HGST} , H_{∇}	Hamiltonian / HGST / driver collision.	Doc II
D (bare dim)	D_n , $\mathfrak{D}$, D_ρ	Dimension / distinguishability / donor collision.	Doc IV
M (bare)	M_n , $\mathbf{M}$, m_i , $\mathcal{M}$	Matrix / mediator / mass / manifold collision.	Doc II
w (5th coord)	$u = \ln(\ell/\ell_0)$	Superseded by logarithmic coordinate in Doc IV.	Doc IV, 2026-05-24
ρ_D (donor)	D_ρ	Looks like density times D .	VSEPR-SIM docs
S (steric)	S_{steric}	Collides with entropy and scale.	VSEPR-SIM docs
Π (bare proj)	$\mathcal{P}_{a \rightarrow b}$ or Π_{ox}	Projection / oxidation plausibility collision.	Doc IV
$P_{a \rightarrow b} : I_S$	$\mathcal{P}_{a \rightarrow b}[\mathfrak{I}_a]$	Old projection form. Implied migration.	Doc II

§15 Open Symbol Questions

These are unresolved questions in the notation system. Each item here blocks finalization of the section it belongs to.

Symbol	Section	Open question
M_n	Scale Ladder (§4)	Mediator, matrix, or metric? Must choose one canonical meaning before Doc V.
η_u	Geometry (§11)	Free parameter in the 5D interval. Needs experimental or theoretical constraint.
k_Q	Geometry (§11)	Sharpness of activation sigmoid. Requires calibration against data for each property.
$\hat{H}_w$	Matrix-Wave (§6)	Dark-scale Hamiltonian. Without this the theory has no new quantitative predictions.
χ_{ab}	Projection (§3)	Is scale compatibility a Boolean gate or a continuous weight? Relation to η_{ab} unclear.

Symbol	Section	Open question
$\mathfrak{A}$	Identity (§5)	Anchor role: weighted mean of components, or something stronger (e.g. attractor in identity space)?
$\mathbf{T}_{AB}$	Entanglement (§7)	Space defined ($\{-1, 0, +1\}^m$), but dynamics of $\mathbf{T}_{AB}$ over time are not yet defined.
Γ_{AB} vs κ_{AB}	Entanglement (§7)	Both are BLUE . Decide whether both are needed or one subsumes the other.
$D_{\text{eff}}(\ell)$ functional form	Geometry (§11)	$a_i(\ell)$ curves are defined abstractly. Explicit functional form from first principles needed.

§16 Promotion and Demotion Log

This log is updated whenever a symbol changes status. Most recent entry first.

Date	Symbol	From	To	Reason
2026-05-24	$u = \ln(\ell/\ell_0)$	—	GREEN	Introduced in Doc IV as canonical logarithmic scale coordinate. Supersedes bare w .
2026-05-24	w	YELLOW	RED	Replaced by u in Doc IV.
2026-05-24	$\Pi_{m \leftarrow n}$	BLUE	GREEN	Accepted as Doc IV projection notation. Equivalent to $\mathcal{P}_{n \rightarrow m}$.
2026-05-24	$A_Q(\ell)$	—	GREEN	Activation function introduced Doc IV. Sigmoid form accepted.
2026-05-24	$\mathbb{C}_p$	—	GREEN	Complex fibre introduced Doc IV. Fibre bundle structure accepted.
2026-05-24	ds_5^2	BLUE	GREEN	5D interval fully formalised in Doc IV.
2026-05-24	I (plain)	YELLOW	RED	Replaced by $\mathfrak{I}$ and $\mathfrak{D}$.
2026-05-24	$\mathcal{P}_{a \rightarrow b}$	BLUE	GREEN	Accepted as canonical projection operator.
2026-05-24	S (bare)	YELLOW	ORANGE	Collision confirmed: scale S_n vs entropy S vs steric S_{steric} .
2026-05-24	χ_{ab}	—	BLUE	Candidate scale-access compatibility variable introduced.
2026-05-24	η_{ab}	—	BLUE	Candidate projection efficiency variable introduced.
2026-05-24	ϵ_{ab}	—	BLUE	Candidate projection error variable introduced.
2026-05-24	Γ_{AB}	—	BLUE	Candidate relation-graph symbol introduced.
2026-05-24	κ_{AB}	—	BLUE	Candidate coupling-strength scalar introduced.
— add new entries above this line, most recent first —				

IKK Notation Registry — VSEPR-SIM Theoretical Division. A symbol without a status tag is a disaster waiting for an audience.